Questions

Q1.

Two small children, Ajaz and Beth, are running a 100 m race along a straight horizontal track.

They both start from rest, leaving the start line at the same time.

Ajaz accelerates at 0.8 m s^{-2} up to a speed of 4 m s^{-1} and then maintains this speed until he crosses the finish line.

Beth accelerates at 1 m s^{-2} for T seconds and then maintains a constant speed until she crosses the finish line.

Ajaz and Beth cross the finish line at the same time.

(a) Sketch, on the same axes, a speed-time graph for each child, from the instant when they leave the start line to the instant when they cross the finish line.

(3)

(b) Find the time taken by Ajaz to complete the race.

(4)

(c) Find the value of T

(4)

(d) Find the difference in the speeds of the two children as they cross the finish line.

(2)

(Total for question = 13 marks)

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(**Q07 WME01/01, June 2022**)

Q2.

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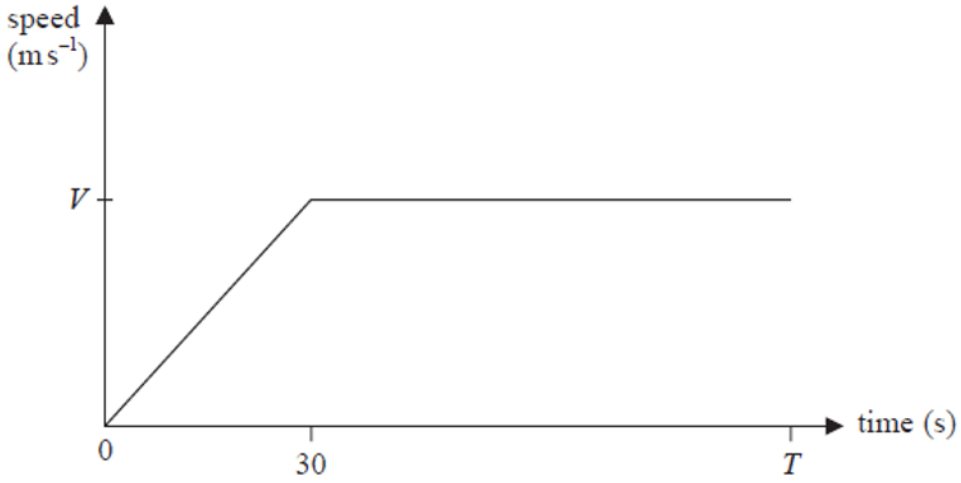


Figure 2

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The speed-time graph in Figure 2 illustrates the motion of a car travelling along a straight horizontal road.

At time $t = 0$, the car starts from rest and accelerates uniformly for 30 s until it reaches a speed of V m s^{-1}

The car then travels at a constant speed of V m s^{-1} until time $t = T$ seconds.

(a) Show that the distance travelled by the car between $t = 0$ and $t = T$ seconds is $V(T - 15)$ metres.

(2)

A motorbike also travels along the same road.

- The motorbike starts from rest at time ***t* = 10 s** and accelerates uniformly for 40 s
- The acceleration of the motorbike is the **same** as the acceleration of the car
- The motorbike then travels at a constant speed for a further 10 s before decelerating uniformly until it reaches a speed of V m s^{-1} at time T seconds

(b) On Figure 2, sketch a speed-time graph for the motion of the motorbike.

[If you need to redraw your sketch, there is a copy of Figure 2.]

(2)

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(c) Show that the constant speed of the motorbike is $\frac{4V}{3}$ m s^{-1}

(2)

At time $t = T$ seconds, the distance travelled by each vehicle is the same.

(d) Find the value of T

(5)

Only use this copy of Figure 2 if you need to redraw your sketch.

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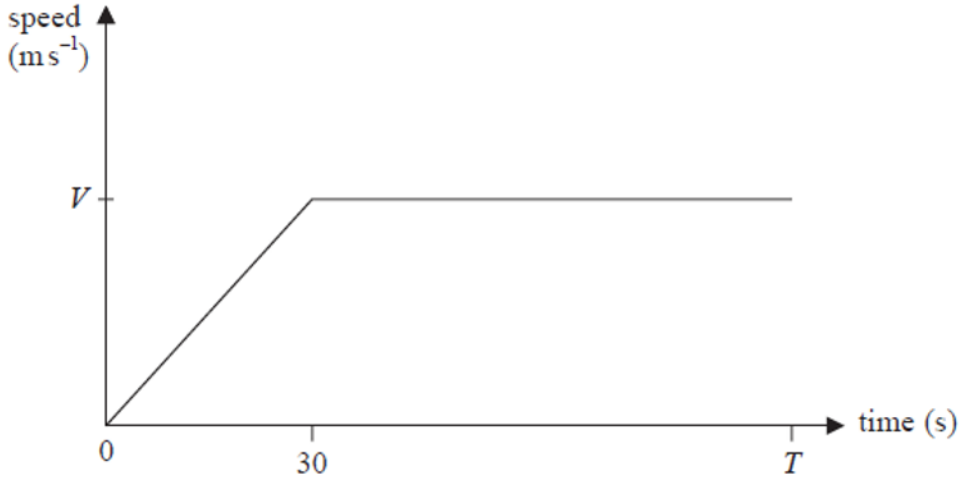


Figure 2

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(Total for question = 11 marks)

(Q05 WME01/01, June 2023)

Q3.

A train travels along a straight horizontal track between two stations *A* and *B*.

The train starts from rest at station *A* and accelerates uniformly for *T* seconds until it reaches a speed of 20 m s^{-1}

The train then travels at a constant speed of 20 m s^{-1} for 3 minutes before decelerating uniformly until it comes to rest at station *B*.

The magnitude of the acceleration of the train is twice the magnitude of the deceleration.

(a) On the axes below, sketch a speed–time graph to illustrate the motion of the train as it moves from
station *A* to station *B*.

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If you need to redraw your graph, use the axes below RocketRevise

(3)

Stations *A* and *B* are 4.8 km apart.

(b) Find the value of *T*

(5)

(c) Find the acceleration of the train during the first *T* seconds of its motion.

(2)

(Total for question = 10 marks)



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(Q01 WME01/01, Jan 2023)

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Q4.

Two trams, tram *A* and tram *B*, run on parallel straight horizontal tracks. Initially the two trams are at rest in the depot and level with each other.

At time $t = 0$, tram *A* starts to move. Tram *A* moves with constant acceleration 2 ms $^{-2}$ for 5 seconds and then continues to move along the track at constant speed.

At time $t = 20$ seconds, tram *B* starts from rest and moves in the same direction as tram *A*. Tram *B* moves with constant acceleration 3 ms $^{-2}$ for 4 seconds and then continues to move along the track at constant speed.

The trams are modelled as particles.

(a) Sketch, on the same axes, a speed-time graph for the motion of tram *A* and a speed-time graph for the motion of tram *B*, from $t = 0$ to the instant when tram *B* overtakes tram *A*.

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(3)

At the instant when the two trams are moving with the same speed, tram *A* is *d* metres in front of tram *B*.

(b) Find the value of *d*.

(5)

(c) Find the distance of the trams from the depot at the instant when tram *B* overtakes tram *A*.

(5)

(Total for question = 13 marks)

(**Q08 WME01/01, June 2021**)

Q5.

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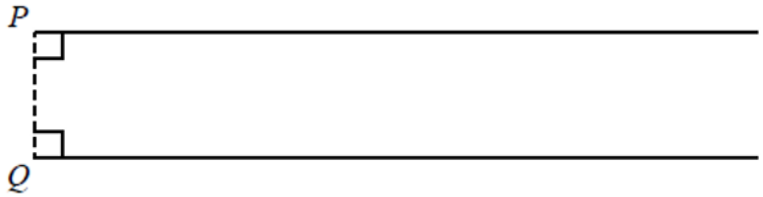


Figure 2

Two cars, *A* and *B*, move on parallel straight horizontal tracks. Initially *A* and *B* are both at rest with A at the point *P* and *B* at the point *Q*, as shown in Figure 2. At time $t = 0$ seconds, A starts to move with constant acceleration a m s^{-2} for 3.5 s, reaching a speed of 14 m s^{-1}. Car A then moves with constant speed 14 m s^{-1}.

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(a) Find the value of a.

(2)

Car B also starts to move at time $t = 0$ seconds, in the same direction as car A. Car B moves with a constant acceleration of 3 m s^{-2}. At time $t = T$ seconds, B overtakes A. At this instant A is moving with constant speed.

(b) On a diagram, sketch, on the same axes, a speed-time graph for the motion of A for the interval $0 \leq t \leq T$ and a speed-time graph for the motion of B for the interval $0 \leq t \leq T$.

(3)

(c) Find the value of T.

(8)

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(d) Find the distance of car B from the point Q when B overtakes A.

(1)

(e) On a new diagram, sketch, on the same axes, an acceleration-time graph for the motion of A for the interval $0 \leq t \leq T$ and an acceleration-time graph for the motion of B for the interval $0 \leq t \leq T$.

(3)

(Total for question = 17 marks)

(**Q06 WME11/01, Specimen papers**)

Q6.

At time $t = 0$, a small ball is projected vertically upwards from a point A which is 24.5 m above the ground. The ball first comes to instantaneous rest at the point B, where $AB = 19.6$ m and first hits the ground at time $t = T$ seconds.

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The ball is modelled as a particle moving freely under gravity.

(a) Find the value of T.

(6)

(b) Sketch a speed-time graph for the motion of the ball from$t = 0$ to $t = T$ seconds.

(No further calculations are needed in order to draw this sketch.)

(2)

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(Total for question = 8 marks)

(Q04 WME01/01, Jan 2022)

Q7.

A small ball is projected vertically upwards with speed 29.4 m s^{-1} from a point A which is 19.6 m above horizontal ground.

The ball is modelled as a particle moving freely under gravity until it hits the ground. It is assumed that the ball does not rebound.

(a) Find the distance travelled by the ball while its speed is less than 14.7 m s^{-1}

(3)

(b) Find the time for which the ball is moving with a speed of more than 29.4 m s^{-1}

(3)

(c) Sketch a speed-time graph for the motion of the ball from the instant when it is projected from *A* to the instant when it hits the ground. Show clearly where your graph meets the axes.

(3)

(Total for question = 9 marks)

(Q05 WME01/01, Oct 2022)

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Q8.

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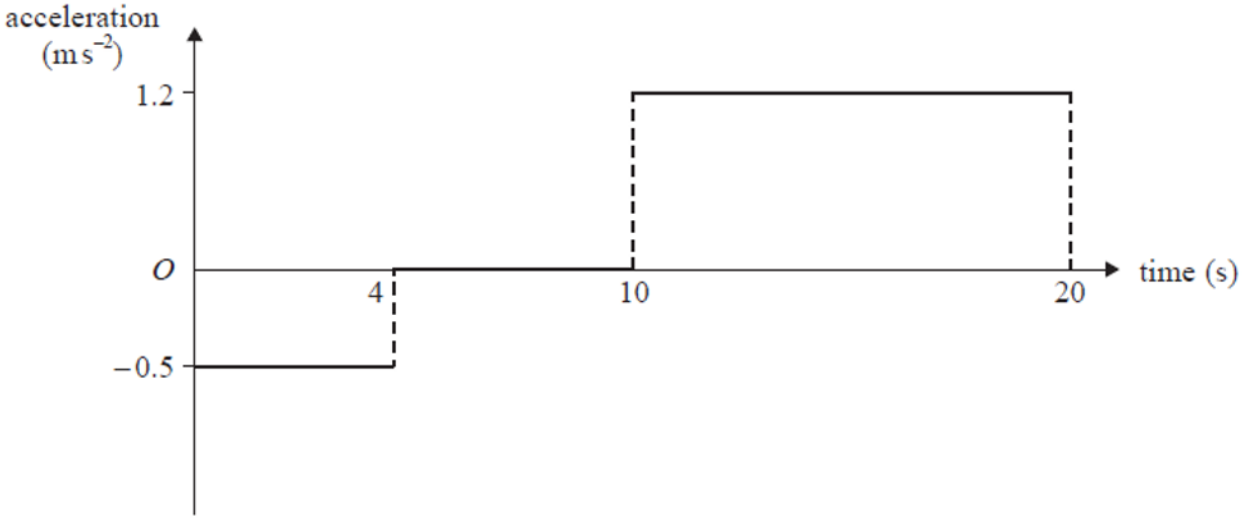


Figure 2

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Two fixed points, A and B, are on a straight horizontal road.

The **acceleration-time** graph in Figure 2 represents the motion of a car travelling along the road as it moves from A to B.

At time $t = 0$, the car passes through A with speed 8 m s^{-1}

At time $t = 20$ s, the car passes through B with speed v m s^{-1}

(a) Show that $v = 18$

(3)

(b) Sketch a speed-time graph for the motion of the car from A to B.

(3)

(c) Find the distance AB.

(4)

(Total for question = 10 marks)

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(Q02 WME01/01, Oct 2023)

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